1. **Problem + Algorithm**

Experiment No.: 14

Title: Gram Matrix of a Dataset Using Different Kernels

Aim:
Compute and visualise the Gram matrix of a dataset using different kernel functions and check its properties.

Problem Statement:
For 80 samples of the two-moons dataset (noise 0.08, random seed 1, 40 samples per class), the Gram matrix $K[i, j] = k(x_i, x_j)$ is computed for six kernels: linear $x \cdot y$, polynomial $(x \cdot y + 1)^3$, RBF $\exp(-\gamma\|x - y\|^2)$ with $\gamma = 1$ and $\gamma = 10$, Laplacian $\exp(-\gamma\|x - y\|_1)$ with $\gamma = 1$, and sigmoid $\tanh(0.5 \cdot x \cdot y)$. A valid kernel gives a symmetric, positive semi-definite (PSD) Gram matrix, i.e. one with no negative eigenvalues. The samples are sorted by class so that class structure appears as blocks in the heatmaps.

Algorithm:

- Generate 80 two-moons samples and sort them by class.
- Compute the Gram matrix for the six kernels with the scikit-learn pairwise kernel functions.
- Check the RBF matrix against a manual computation using $\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2 \cdot x \cdot y$.
- Check symmetry: $K = K^T$.
- Compute the smallest eigenvalue of each K; the matrix is PSD if it is at least -10^{-8} (a tolerance for rounding error).
- Display each Gram matrix as a heatmap, with white lines separating class 0 from class 1.

2. **Program**

Important sections of the program (**q14_gram_matrix.py**):

Data and Gram matrices

```
X, y = make_moons(n_samples=80, noise=0.08, random_state=1)
order = np.argsort(y)
X, y = X[order], y[order]                  # sorted by class

kernels = {
    "Linear: x·y":          linear_kernel(X),
    "Polynomial (deg=3)":   polynomial_kernel(X, degree=3,
                                              gamma=1, coef0=1),
    "RBF (gamma=1)":        rbf_kernel(X, gamma=1.0),
    "RBF (gamma=10)":       rbf_kernel(X, gamma=10.0),
    "Laplacian (gamma=1)":  laplacian_kernel(X, gamma=1.0),
    "Sigmoid (tanh)":       sigmoid_kernel(X, gamma=0.5, coef0=0),
}
```

Core algorithm (manual check and properties)

```
sq = (np.sum(X**2, 1)[:, None] + np.sum(X**2, 1)[None, :]
      - 2 * X @ X.T)
assert np.allclose(np.exp(-1.0 * sq), kernels["RBF (gamma=1)"])

for name, K in kernels.items():
    min_eig = np.linalg.eigvalsh(K).min()
    symmetric = np.allclose(K, K.T)
```

```
    valid = min_eig > -1e-8                # PSD check
```

The complete executable program can be maintained digitally through Github.

3. **Output + Analysis**

Objective Function:
Compute K[i, j] = k(x_i, x_j) for each kernel and test whether K is symmetric and positive semi-definite.

Dataset: Two moons, 80 samples (40 per class), noise = 0.08

Kernels: Linear, polynomial (degree 3), RBF (γ = 1, 10), Laplacian (γ = 1), sigmoid (γ = 0.5)

PSD Tolerance: -10^{-8} on the smallest eigenvalue

Random Seed: 1

Program Execution
Final Optimal Solution:
Five of the six kernels give a valid (PSD) Gram matrix; the sigmoid kernel does not

Smallest Eigenvalue of the Sigmoid Matrix:
−2.2389

Execution Status: Successfully Executed

Manual RBF computation matches rbf_kernel (assert passed)
Linear: x·y : Symmetric = True, Min eigenvalue = -0.0000, Valid (PSD) = True
Polynomial (deg=3) : Symmetric = True, Min eigenvalue = -0.0000, Valid (PSD) = True
RBF (gamma=1) : Symmetric = True, Min eigenvalue = 0.0000, Valid (PSD) = True
RBF (gamma=10) : Symmetric = True, Min eigenvalue = 0.0000, Valid (PSD) = True
Laplacian (gamma=1) : Symmetric = True, Min eigenvalue = 0.0077, Valid (PSD) = True
Sigmoid (tanh) : Symmetric = True, Min eigenvalue = -2.2389, Valid (PSD) = False

Gram Matrix Heatmaps

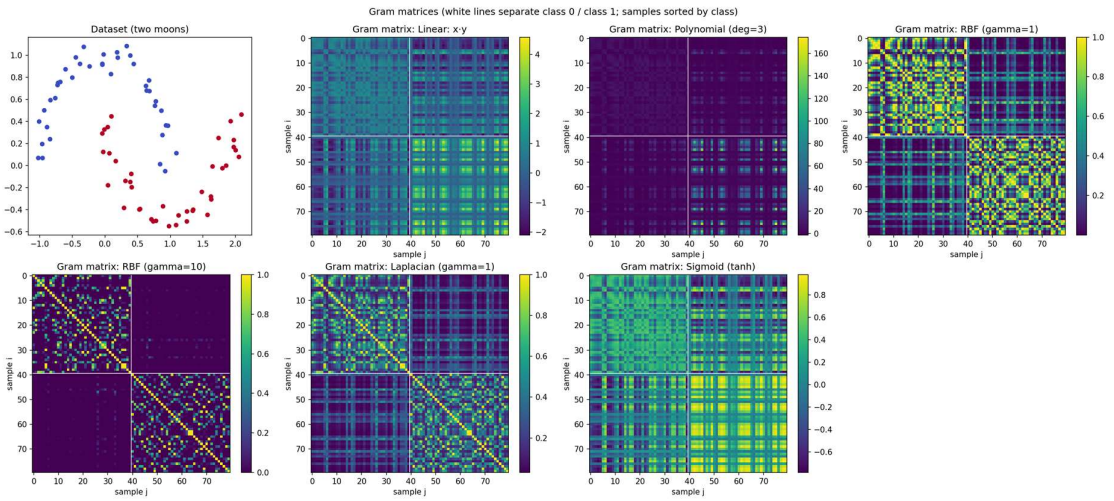


Figure: Gram Matrices of the Two-Moons Data for Different Kernels

- X-axis → Sample j
- Y-axis → Sample i (colour = k(x_i, x_j); white lines separate class 0 and class 1)

The samples are sorted by class, so a kernel that captures the class structure shows two bright diagonal blocks (class 0 and class 1) and dark off-diagonal blocks. This is clearest for RBF with $\gamma = 10$, where the matrix is almost zero between the classes and bright only close to the diagonal, followed by RBF with $\gamma = 1$ and the Laplacian kernel. The linear, polynomial and sigmoid matrices show a striped pattern instead of blocks, because their values depend on the length and direction of the vectors (the polynomial values reach about 170) rather than on the distance between samples. The sigmoid kernel also contains negative values and is the only matrix that is not positive semi-definite.

Parameter Modification

S. No.	Parameter	Original Value	Modified Value	Original Result	Modified Result	Observation
1	RBF γ	1	10	Min eigenvalue ≈ 0 (PSD) Mean off-diagonal = 0.3405	Min eigenvalue ≈ 0 (PSD) Mean off-diagonal = 0.0671	Larger γ makes similarity more local, so K moves towards the identity matrix
2	RBF γ	1	0.1	Min eigenvalue ≈ 0 (PSD) Mean off-diagonal = 0.3405	Min eigenvalue ≈ 0 (PSD) Mean off-diagonal = 0.8313	Smaller γ makes almost all points look similar, so K approaches the all-ones matrix
3	Polynomial degree	3	2	Min eigenvalue ≈ 0 (PSD) Mean off-diagonal = 6.6904	Min eigenvalue ≈ 0 (PSD) Mean off-diagonal = 2.6732	Lower degree gives smaller kernel values and the matrix stays PSD
4	Sigmoid γ	0.5	0.1	Min eigenvalue = −2.2389 Mean off-diagonal = 0.1124	Min eigenvalue = −0.0489 Mean off-diagonal = 0.0289	The violation is much smaller but the matrix is still not PSD, so the sigmoid kernel is not valid here

Conclusion: The linear, polynomial, RBF and Laplacian Gram matrices are symmetric and positive semi-definite, so they are valid kernels. The sigmoid kernel ($\gamma = 0.5$, coef0 = 0) is not PSD (smallest eigenvalue −2.2389), so it does not correspond to an inner product in a feature space for these parameters. The RBF parameter γ controls how local the similarity is, and the manual RBF computation matches the library result. Results are from a run with random seed 1.